# No. of Functional Features Included in the Solution

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|---|---|
| **Date** | 03 October 2026 |
| **Team ID** | 2 |
| **Project Name** | FitBuddy – AI Fitness Plan Generator using Gemini Models |
| **Maximum Marks** | 5 Marks |

**Functional Features Overview:**

| S.No | Feature Name | Feature Description | Module / Component | Status | Marks Contribution |
|---|---|---|---|---|---|
| 1 | Personalized plan generation | 7-day plan from age, weight, goal, intensity and experience; validated JSON | gemini_generator.py, routes.py | Done | Core |
| 2 | Day-card plan display | Warm-up, exercises (sets × reps, rest, form cue), cooldown; rest days highlighted; print | result.html, _plan.html | Done | Core |
| 3 | Nutrition / recovery tip | 2–3 sentence goal-based tip; built-in fallback tip | gemini_flash_generator.py | Done | Core |
| 4 | Feedback-based revision | Revises the latest version from plain-language feedback (≤500 chars, delimited) | updated_plan.py | Done | Core |
| 5 | Version history | Every revision stored; view any earlier version | database.py, result.html | Done | Core |
| 6 | Admin dashboard | Password login; all users; original vs latest plan; delete user | all_users.html, admin_login.html | Done | Core |
| 7 | JSON API + Swagger | Generate, tip, plan, update, history and health endpoints at /docs | routes.py, schemas.py | Done | Additional |
| 8 | Reliability features | Retry, fallback model, demo mode, friendly error page, nothing saved on failure | gemini_client.py, demo_data.py, error.html | Done | Additional |

**Feature Summary:**

| Metric | Count / Value |
|---|---|
| Total Features Planned | 8 |
| Total Features Implemented | 8 |
| Core / Must-Have Features | 6 |
| Additional / Nice-to-Have Features | 2 |
| Features Tested & Verified | 8 (245 automated tests + manual real-Gemini run) |

**Feature Category Breakdown:**

| S.No | Category | Features in Category | Example Features |
|---|---|---|---|
| 1 | User Interface (UI) | 3 | Profile form with validation and loading state; day cards; version history and print |
| 2 | Backend / Logic | 3 | Plan generation, feedback revision, tip generation |
| 3 | Database / Storage | 2 | Versioned plans; cascade delete |
| 4 | API / Integration | 2 | Gemini structured output with fallback; JSON API with Swagger |
| 5 | Security / Authentication | 2 | Admin password + signed cookie; input validation and prompt-injection guard |

Some features span more than one category, so category counts add up to more than 8.